Arata Ishikawa

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Education

University of California, Berkeley Aug. 2023 – May 2027
Bachelor of Science, Engineering Physics – GPA 3.830/4.000 *Berkeley, CA*

Relevant Coursework: Analytical Mechanics, Real Analysis, Numerical Analysis (In Progress), Machine Learning, Fluid Mechanics, Statistical Mechanics, Optimization Models, Quantum Mechanics, Electromagnetism and Optics

Research Experience

Moncrief Undergraduate Research Intern Jun. 2026 – Aug. 2026
Oden Institute for CSEM, UT Austin – PI: Professor Michael Sacks *Austin, TX*

- Fitted human, murine, and ovine heart models with GPU (JAX) accelerated B-spline surface and volume fitting algorithms
- Verified fitting pipeline accuracy (up to order 1e-1 error) with pointwise strain comparisons to synthetic FE myocardium mesh models

Undergraduate Research Assistant Sep. 2025 – Present
Space Sciences Laboratory – PI: Dr. Mike Wong, Dr. Philip Marcus *Berkeley, CA*

- Developed and compared multiple deep learning workflows and integrated into existing ACCIV software to accelerate velocity field extraction
- Utilized Advection Corrected Correlation Image Velocimetry (ACCIV) to estimate velocity fields of planetary atmospheres such as Jupiter and Saturn.
- Analyzed satellite and space telescope (JWST, Hubble) images of Jupiter's Great Red Spot and Oval BA for vorticity patterns

Undergraduate Research Assistant Jun. 2025 – Jan. 2026
FLOW Lab – PI: Professor Simo Makiharju *Berkeley, CA*

- Created tomographic particle tracking velocimetry algorithm that calculates particle coordinates from a series of projections based on the governing physical equations
- Designed and tested x-ray velocimetry and limited angle tracking of tracer particles with synthetic data + noise (up to 1e-4 error)

Undergraduate Research Assistant Aug. 2024 – May 2025
Yildiz Lab – PI: Professor Ahmet Yildiz *Berkeley, CA*

- Performed various biochemistry techniques involved in designing, producing, and purifying motor proteins kinesin, dynein and cargo adaptor protein HAP1
- Executed various experimental techniques such as TIRF Microscopy, gel electrophoresis, insect cell protein expression, protein purification with column chromatography for single molecule imaging and analysis

Conference Poster Presentations

Machine Learning Methods for Analysis of Jupiter Atmospheres (Upcoming) Dec. 2026
American Geophysical Union – accepted *San Francisco, CA*

3D Kinematic Study of the Murine Heart Oct. 2026
Biomedical Engineering Society – accepted *Orlando, FL*

3D Kinematic Study of the Murine Heart Aug. 2026
Oden Institute for Computational Science, Engineering and Mathematics *Austin, TX*

Additional Experiences

Physics Course Tutor and Reader Aug. 2025 – Present
Physics Department *Berkeley, CA*
- Graded problem sets, discussion worksheets, lab worksheets, and finals for over 300+ students in 1 year
- Led office hours and tutoring sessions for introductory electromagnetism and optics course
- Hosted review sessions with personally designed review worksheets covering course content

Software Engineering Intern June 2023 – July 2023
Roku Inc. *San Jose, CA*
- Programmed and developed RokuTV app using BrightScript/SceneGraph
- Composed and presented business plan alongside application to senior engineers and project managers

Software Engineering Intern June 2022 – July 2022
Intuit Inc. *San Jose, CA*
- Utilized TypeScript/React/React Native to handle API calls to advance front-end development
- Showcased and presented final product to the lead engineering team at Intuit

Honors and Awards

Dean's List SP24, FA24
College of Engineering, College of Letters and Science *Berkeley, CA*
- A semester GPA in the top 10 percent of all undergraduates within college

Honors to Date FA24, SP25, FA25, SP26
College of Engineering, College of Letters and Science *Berkeley, CA*
- Have achieved an overall grade point average equal to or greater than the grade point average in the top 20 percent of all undergraduates within college

Community Engagement and Extracurriculars

Volunteering Health Interpreter's Organization September 2023 – Present
Japanese Health Interpreter, Finance Director *Berkeley, CA*
- Served as Japanese Department Coordinator, organizing and leading groups of bilingual health interpreters for various healthcare tasks to serve the local Japanese community
- Took on the role of Finance director, managing club finances and overseeing collaborations with local businesses for fundraising purposes
- Undertook medical and non-medical Japanese translations/interpretation sessions as part of a student-led non-profit organization

Nikkei Student Union September 2023 – Present
Community Service and Issues Staff *Berkeley, CA*
- Organized semesterly community visits to local senior centers and assembled gift baskets for the seniors
- Contributed to the founding (and ongoing) annual Matsuri festival on Berkeley campus with 100+ attendees

Space Enterprises at Berkeley May 2025 – August 2025
Simulations Member *Berkeley, CA*
- Used ANSYS Fluent to analyze current launch lug fluid flow mechanics
- Researched and designed new launch lugs for better aerodynamic performance using Fusion 360 CAD

Suitcase Clinic May 2025 – Present
Haircutting Clinic *Berkeley, CA*
- Provided haircutting services to the unhoused communities in the local Berkeley area
- Connected with the Berkeley community through outreach and distributed basic resources such as food, hygiene kits, and clothes

Technical Skills

Computer: Python, MATLAB, Linux/Unix environments, JAX, ParaView, R, LaTeX, SQL, HTML/CSS, Git, SciPy, NumPy, Matplotlib
Engineering: ANSYS Fluent, ANSYS Mechanical, ANSYS Workbench, Fusion 360 CAD
Laboratory: Protein Purification, Cell Protein Expression, TIRF Microscopy, Gel Electrophoresis, PCR, Plasmid Design